LOCALIZATION OF AN ADDITIVE CATEGORY AT A MULTIPLICATIVE SYSTEM IS ADDITIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; it is terse but each ingredient of additivity is accounted for, one from each cited lemma, and no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The preadditive structure on the localization. Same chapter, Lemma lemma-localization-preadditive. Supplies that the localization of a preadditive category at a left or right multiplicative system carries a preadditive structure for which the localization functor is additive.
- Additive functors preserve zero objects and direct sums. Same chapter, Lemma lemma-additive-additive. Supplies that an additive functor transforms zero objects to zero objects and direct sums to direct sums, which transports the additive structure of the source to the localization.

Source reference. Original lemma 05QD: sources/stacks-homology/pinned-original-source.tex, source lines 1971–1971. Navigation pointer invoked at proof line 2091; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0105: sources/stacks-homology/pinned-original-source.tex, source lines 390–390. Navigation pointer invoked at proof line 2093; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ be an additive category. Let S be a left or right multiplicative system. Then $S^{-1}\mathcal{C}$ is an additive category and the localization functor $Q : \mathcal{C} \to S^{-1}\mathcal{C}$ is additive.*

Proof. By Lemma 05QD we see that $S^{-1}\mathcal{C}$ is preadditive and that Q is additive. By Lemma 0105 we conclude that $S^{-1}\mathcal{C}$ has a zero object and direct sums. □